

Basic Mathematics

Mathematics

Lecture Notes

A friendly path through algebra, geometry, and calculus.

June 6, 2026

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Chapter 1

Introduction

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognizes the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, how to formulate precise questions, how to distinguish an object from its notation, how to notice structure, and how to build simple but meaningful chains of reasoning. A solved problem is not valuable only because it produces an answer. It is valuable because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course will focus on three fundamental parts of mathematics: linear algebra, analytic geometry, and calculus. Linear algebra introduces vectors, matrices, systems of equations, and transformations. Analytic geometry shows how algebraic equations describe points, lines, planes, directions, distances, and angles. Calculus gives a language for functions, limits, continuity, derivatives, integrals, change, and accumulation. These topics are not isolated chapters placed next to one another. Together they form a basic language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorize, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as the result of asking a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some arguments will deliberately return in different places. The same habit of asking what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. The same mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognize patterns of reasoning rather than only patterns of exercises.

The number of solved tasks is not the main measure of progress. What matters is how the tasks are solved. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each step was allowed? Did we pause to interpret the result? Did the calculation teach us something about the idea behind it? These questions are part of the work. They are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics

as an art of asking good questions and giving well-reasoned answers.

Chapter 2

Matrices

2.1 What matrices are

A matrix is a rectangular arrangement of mathematical objects, usually numbers. This description is correct, but it is not yet very informative. If we stop at this description, a matrix looks like a table, and matrix algebra looks like a collection of rules for moving numbers around. In this chapter we will try to see more than that. A matrix is a way of organizing information. It can describe data, relations between quantities, coefficients of equations, or a rule that transforms one vector into another.

The first skill is therefore not calculation, but reading. When we see a matrix, we should ask what its shape is, what its entries represent, and how its rows and columns are meant to be interpreted. The same array of numbers may be only a table in one problem, but in another problem it may describe a transformation, a system of equations, or a geometric operation.

Definition

A matrix with m rows and n columns is called an $m \times n$ matrix. It has the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The entry a_{ij} is located in row i and column j .

This notation is simple, but it contains an important discipline. The first index tells us where to look vertically, and the second index tells us where to look horizontally. A student who confuses these indices will often make mistakes later when multiplying matrices or describing systems of equations.

Example

Consider the matrix

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}.$$

This is a 2×3 matrix. The entry m_{11} is 4, the entry m_{12} is -2 , and the entry m_{23} is 5. We are not doing any operation yet. We are only learning to read the object.

The size of a matrix is part of its identity. A row matrix, a column matrix, and a square matrix may contain similar numbers, but they play different roles. A column matrix can represent a vector. A square matrix can sometimes represent a transformation from a space to itself. A rectangular matrix may describe a relation between two collections of quantities of different sizes.

Definition

A matrix is called square if it has the same number of rows and columns. An $n \times n$ matrix is a square matrix of order n .

We will often begin with square matrices because many basic ideas are easiest to see there. However, rectangular matrices are just as important. The point is not to memorize a list of shapes, but to develop the habit of asking: what kind of object am I holding, and what operations make sense for this object?

2.2 Basic matrix operations

Once we know how to read a matrix, we can begin to operate on it. The simplest operations are addition, subtraction, and multiplication by a number. These operations are deliberately simple because they preserve the shape of the matrix. They do not change a table into a different kind of object; they change its entries while keeping the same positions.

Definition

Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same size $m \times n$. Their sum is the matrix

$$A + B = (a_{ij} + b_{ij}),$$

and their difference is the matrix

$$A - B = (a_{ij} - b_{ij}).$$

If λ is a number, then

$$\lambda A = (\lambda a_{ij}).$$

The phrase “of the same size” is not a technical detail. It is the reason the operation is meaningful. When two matrices have the same size, each entry of one matrix has a partner in the other matrix. Addition and subtraction then happen position by position.

Example

Let

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 2+5 & -1+4 \\ 0+(-2) & 3+1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix}.$$

Also,

$$A - B = \begin{pmatrix} 2-5 & -1-4 \\ 0-(-2) & 3-1 \end{pmatrix} = \begin{pmatrix} -3 & -5 \\ 2 & 2 \end{pmatrix}.$$

Finally,

$$-3A = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}.$$

The calculation is not difficult. The important thing is the discipline of positions. We do not add rows to columns, and we do not rearrange entries because it looks convenient. We respect the shape of the object.

These operations satisfy the familiar rules one expects from ordinary arithmetic. For matrices of the same size,

$$A + B = B + A, \quad (A + B) + C = A + (B + C).$$

This should not be surprising: addition is performed entry by entry, and ordinary addition of numbers is commutative and associative.

Remark

Whenever a matrix operation is defined entry by entry, many properties of ordinary arithmetic pass directly to matrices. But this principle must be used carefully. Matrix multiplication will not be entry-by-entry multiplication, and therefore it will behave differently.

2.3 Matrix multiplication

Matrix multiplication is the first operation in this chapter that is not simply performed position by position. This is why it deserves careful attention. Many mistakes with matrices come from treating multiplication as if it were only another version of addition. It is not. Addition compares matching positions. Multiplication combines rows of the first matrix with columns of the second matrix.

Definition

Let $A = (a_{ij})$ be an $m \times n$ matrix and let $B = (b_{ij})$ be an $n \times p$ matrix. The product AB is the $m \times p$ matrix $C = (c_{ij})$, where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

In words, the entry c_{ij} is obtained by multiplying row i of A by column j of B .

The condition on sizes is built into the definition. If A has size $m \times n$, then each row of A has n entries. To multiply this row by a column of B , each column of B must also have n entries. Therefore B must have n rows. We summarize this by writing

$$(m \times n)(n \times p) = m \times p.$$

The inner dimensions must match; the outer dimensions describe the size of the result.

Example

Let

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}.$$

The first entry of AB comes from the first row of A and the first column of B :

$$1 \cdot 2 + 3 \cdot 5 = 17.$$

The first row and second column give

$$1 \cdot 0 + 3 \cdot (-1) = -3.$$

Continuing in the same way,

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}.$$

The example is small, but the point is general. Each entry of the product answers one local question: how does a chosen row of the first matrix interact with a chosen column of the second matrix?

Why the order matters

For ordinary numbers, $ab = ba$. For matrices this is not generally true. The reason is conceptual, not only computational. The product AB asks rows of A to interact with columns of B . The product BA asks a different question.

Using the matrices from the previous example,

$$BA = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}.$$

Thus $AB \neq BA$. The matrices have not changed, but the order has changed, and the question asked by the multiplication has changed.

Remark

The statement $AB \neq BA$ does not mean that two matrices can never commute. Some matrices do commute. The point is that commutativity is not automatic. If it matters, it must be checked or justified from the structure of the matrices.

Matrix multiplication is associative:

$$(AB)C = A(BC),$$

whenever the products are defined. This means that if several compatible matrices are multiplied in a fixed order, parentheses do not change the result. But the order of the matrices themselves still matters. Associativity tells us how to group operations. It does not allow us to swap them.

2.4 Special matrices and visible structure

Some matrices are important because their structure is easy to see. They help us learn a useful lesson: a matrix is not only a collection of entries. The position of the entries matters, and certain patterns can make an operation simple or meaningful.

The zero matrix and the identity matrix

The zero matrix is the matrix whose entries are all zero. It behaves like zero for matrix addition. If A is an $m \times n$ matrix and 0 is the zero matrix of the same size, then

$$A + 0 = A.$$

The identity matrix plays a similar role for multiplication, but only for square matrices.

Definition

The $n \times n$ identity matrix is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

For compatible matrices,

$$I_n A = A, \quad A I_n = A.$$

The identity matrix is important because it represents doing nothing. This phrase is informal, but helpful. Multiplying by the identity preserves the object, just as multiplying a number by 1 preserves the number.

Diagonal matrices

A diagonal matrix is a square matrix whose non-zero entries, if any, lie only on the main diagonal.

Definition

A square matrix $D = (d_{ij})$ is diagonal if

$$d_{ij} = 0 \quad \text{whenever } i \neq j.$$

We often write

$$D = \text{diag}(d_1, d_2, \dots, d_n).$$

Diagonal matrices are useful because they act independently on different positions. There is no mixing between coordinates. This is why many calculations with diagonal matrices are transparent.

Example

Let

$$D = \text{diag}(3, -1, 2).$$

Then

$$D^2 = \text{diag}(9, 1, 4), \quad D^3 = \text{diag}(27, -1, 8).$$

The power is computed by taking powers of the diagonal entries. This works because all off-diagonal entries are zero, so the coordinates do not mix.

If two diagonal matrices have the same size, then they commute. Indeed,

$$\text{diag}(a_1, \dots, a_n) \text{diag}(b_1, \dots, b_n) = \text{diag}(a_1 b_1, \dots, a_n b_n),$$

and the same result is obtained in the opposite order because ordinary numbers commute. This is a good example of structure explaining an algebraic fact.

Patterns worth noticing

When a matrix appears in a problem, it is useful to ask whether it has visible structure. Is it diagonal? Does it contain many zeros? Are two rows equal? Are rows or columns multiples of one another? Does it look symmetric? Such observations can simplify computations, but more importantly they help us understand the object.

Remark

A calculation often becomes easier after we notice structure. But noticing structure is not a trick; it is part of mathematical understanding. We learn not only to compute with objects, but also to see them.

2.5 Row operations and problem solving

Matrices are often used because they allow us to transform a problem without rewriting every sentence of the problem. One of the most important examples is the use of row operations. Row operations will become essential when we solve systems of linear equations, but it is useful to meet them already here.

A row operation changes a matrix in a controlled way. We do not change entries randomly. We perform an operation that has a clear description and a clear purpose.

Definition

The basic elementary row operations are:

- swapping two rows,
- multiplying a row by a non-zero number,
- adding a multiple of one row to another row.

The value of row operations is not that they make matrices look different. Their value is that they help reveal structure. For example, they can help us simplify a matrix, detect whether equations are dependent, or prepare a system for solving.

Example

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}.$$

Suppose we replace the second row by the second row minus twice the first row. Then

$$(2, 4, 3) - 2(1, 2, -1) = (0, 0, 5).$$

The new matrix is

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}.$$

This operation did not merely produce new numbers. It revealed that part of the second row was already explained by the first row, except for the last entry.

This example should be read slowly. The operation has three layers. First, there is arithmetic. Second, there is a transformation of the row. Third, there is interpretation: the new row tells us something about dependence between rows.

Column vectors as matrices

A column vector may be viewed as a matrix with one column. This viewpoint will be used constantly in linear algebra. It allows us to write vector equations and matrix equations in the same language.

For instance,

$$u = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

is a 3×1 matrix. Its transpose is the row matrix

$$u^\top = \begin{pmatrix} 2 & -1 & 4 \end{pmatrix}.$$

The distinction between a row and a column is not cosmetic. The products $u^\top u$ and $uu^\top$ have different sizes and different meanings:

$$u^\top u$$

is a 1×1 matrix, while

$$uu^\top$$

is a 3×3 matrix.

This is another example of a recurring principle: before calculating, read the shapes.

How to approach matrix exercises

When solving matrix exercises, the aim is not to rush toward the answer. The answer is only the last line of a process. A good solution should make the process visible.

Summary

When working with matrices, ask:

- What is the size of each matrix?
- Which operations are defined?
- Am I adding entries position by position, or multiplying rows by columns?
- Does the order of multiplication matter here?
- Is there a visible structure that simplifies the problem?
- What does the result tell me about the original objects?

A student who asks these questions is not merely applying a recipe. They are beginning to use matrices as mathematical objects. That is the real purpose of this chapter.

Chapter 3

Linear Algebra

This chapter is a temporary container for the remaining introductory material related to linear algebra. As the course structure develops, its topics will be promoted into separate chapters.

3.1 Vectors

This section will introduce vectors and their basic operations.

3.2 Systems of Linear Equations

This section will introduce systems of linear equations and their connection with matrices.

Chapter 4

Analytic Geometry

This chapter introduces geometric objects described using coordinates and algebraic equations.

4.1 Coordinates

This section will introduce coordinate systems and the representation of points.

4.2 Lines

This section will introduce equations of lines and their geometric interpretation.

4.3 Planes

This section will introduce equations of planes in three-dimensional space.

Chapter 5

Calculus

This chapter introduces functions, limits, continuity, and derivatives.

5.1 Functions

This section will introduce functions as mathematical objects describing dependence between quantities.

5.2 Limits

This section will introduce the concept of a limit.

5.3 Continuity

This section will introduce continuity and its basic interpretation.

5.4 Derivatives

This section will introduce derivatives as rates of change.